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Abstract

The rapid adoption of generative AI tools such as ChatGPT, Gemini, and Claude in higher education has produced inconsistent findings regarding their impact on student academic performance. Some studies report modest grade improvements of 1 to 1.5 points in AI-permissive courses while others report no significant changes or long-term declines in learning outcomes. This study examines two research questions: RQ1: How does the use of generative AI tools affect GPA and course pass rates among undergraduate STEM students at four-year universities? RQ2: Does this impact differ between online and in-person learning environments? Using a mixed methods design, we surveyed approximately 50 undergraduate students at the University of Washington and conducted semi-structured interviews with ten follow-up participants. Quantitative survey data will be analyzed using descriptive statistics and comparisons across learning formats, while interview transcripts will be thematically coded to capture student perspectives on AI's influence on learning, academic performance, and mental health. We anticipate findings will reflect the mixed nature of existing literature while providing richer insight into student experiences that large-scale quantitative studies have not captured.

Introduction

Generative artificial intelligence tools including ChatGPT, Gemini, and Claude have been widely integrated into higher education. Many college students use these AI tools to brainstorm ideas, summarize the materials they have read, study course information, check grammar, and do their homework. The rapid adoption of generative AI has sparked numerous discussions regarding the role of the mentioned tool in the educational process, especially in relation to STEM where technological tools are highly utilized during courses. There is very limited data regarding how academic performance differs for students who are enrolled in online versus in-person courses because prior research performed before the introduction of AI indicated there were no statistically relevant findings (Cavanaugh & Jacquemin, 2015).

Additionally, it is possible that any perceived differences in academic performance between students who take an online course versus those taking an in-classroom course may not be due exclusively to AI use within a course. The design of each of these types of educational settings and other factors, including instructor participation, peer-to-peer

interactions, and the pace at which the material is covered, will also impact the outcome of each respective setting. In this study, we hope to isolate the effect that AI tools have on students' educational experiences from other aspects of their experience based on the type of course they are enrolled in.

Recent publications indicate that students participating in courses where AI is permitted have expressed increased confidence in completing assigned work and decreased levels of academic-related anxiety than did students who do not participate in AI-permitted courses. However, some researchers note that while AI provides an alternative method for completing many tasks previously completed by hand, students are now questioning the relative importance or value of their skills compared to those provided through an AI application. These changing perspectives suggest that students perceive AI as a means to improve their education rather than simply as a resource to enhance their ability to complete homework assignments, thus potentially causing them to redefine what constitutes academic achievement and self-confidence (Hausman et al., 2025).

Current research findings provide conflicting results about whether the integration of AI helps learners acquire new skills or hinders their academic performance. Some researchers indicate that AI has an impact on the improvement of academic performance, creativity, and learning efficiency, whereas others reveal possible negative consequences of using generative AI such as the development of long-term dependency on technology and lack of ability to think critically (Hu et al., 2025; Wu & Yu, 2024). At the same time, existing literature does not provide sufficient attention to individual experiences of students in relation to how the use of AI affects their academic outcomes. Furthermore, current evidence does not include qualitative insights obtained through interviews that can provide additional knowledge about AI experiences of undergraduate students. Moreover, most studies conducted to compare online and in-person course formats have not considered generative AI tools.

For this reason, this study will focus on exploring how the use of generative AI tools influences the perception of the students' academic performance and whether there are any differences in AI-related perceptions and experiences of students in online and in-person courses. In order to answer this research question, a mixed-methods research methodology will be used. Specifically, surveys will be conducted among 50 undergraduate students enrolled in four-year university classes who will be asked to disclose their use of AI and the ways it impacts their learning outcomes and habits. Furthermore, approximately ten of those students will participate in an additional follow-up interview, which will provide additional contextual detail regarding how AI is used within the context of their undergraduate education. The interview participants are expected to be purposefully selected as opt-in respondents on the survey with variation

based on both frequency of AI use and the method by which they take courses (i.e., online vs. in-person).

The major contributions of this study are: (1) exploring student perceptions of the impact of AI usage on academic performance, (2) comparing AI experiences in online and offline courses, and (3) collecting qualitative data related to students' emotions and learning experiences in connection with generative AI.

Background/Related work

Existing research articles on generative AI in higher education have mainly focused on how students use AI tools and whether those tools improve or hinder academic performance. Through data analysis of surveys, meta-analysis, and quantitative data, they were able to gather information around AI use in education and provide suggestions to educators, institutions, and even students. The results varied from positive to negative, and trends varied across different educational contexts and time periods.

The research literature may be categorized into three connected questions about the relationship between AI and academic achievement as follows: Does the use of AI result in a statistically significant impact on final grades/pass rates? Are the results from using AI positive or negative? Do course delivery methods affect these relationships? Prior to the emergence of AI technology, researchers reported that there was little difference in learning outcomes between students who took classes face-to-face or remotely (Cavanaugh & Jacquemin, 2015; Hoffman & Elmi, 2020); therefore, course delivery method appears to be an ineffective predictor for learning outcomes. However, relatively recently published research reports indicate that many larger studies have found moderate increases in grades when AI technology was permitted during class time (Hausman et al., 2025) and other smaller studies report that such increased grades were dependent upon several factors including the type of tasks completed, the length of time students had been utilizing AI technology and how they utilized AI technology. Overall, a common thread among all research reviewed was that most studies either examined the objective measure of student grades and did not examine the subjective experience of students related to their utilization of AI technology or relied on self-reporting measures of students' experiences with AI technology and did not correlate these responses to actual performance data. Lastly, very few studies examined whether the way AI tools were utilized during remote classes vs. face-to-face classes interacted differently in the post-ChatGPT education environment. The following section will organize and describe previous research based on whether AI resulted in improved grades/achievement prior to addressing the areas of research that this study will address.

Before the widespread adoption of AI, studies comparing academic performances of both online and in-person learning environments found no significant disparity in grade performances before the widespread adoption of AI. One large study found a GPA disparity of less than 0.07 between online and in-class learners, showing that course format had no substantial impact on grades (Cavanaugh & Jacquemin, 2015). Newer studies, however, indicate that the inclusion of AI has changed this relationship. A study examining data from 36,000 students enrolled in 6,000 courses between 2018 and 2024 at a leading Israeli University found that scores in classes that allowed AI usage improved by 1-1.5 points on a 0-100 scale, and the number of students who failed decreased by one-third (Hausman et al., 2025), suggesting that AI availability is now a meaningful factor in academic performance.

No Effect of AI on Student Performance

Research looking into the usage and perceptions of generative AI technology among college students found that out of 634 students from Middlebury College between December 2024 and February 2025, only 41.1% of students found improvements in their grades as measured by self-reported GPA changes, while 55.4% reported no significant changes, and 3.5% found that their grades declined with the use of AI (Contractor & Reyes, 2025).

A comparison of scores from quizzes, midterms and finals for all students enrolled in a graduate level introductory biostatistics course offered in both on-line and face-to-face environments from 2013 through 2018 also found that students who took courses in an on-line format scored approximately 2.5 % less than those taking their classes in person on quizzes; however, they scored about 0.9 % more on their final exams (Hoffman & Elmi, 2020). When looking at factors beyond just the method of delivery (such as GPA and GRE test scores) it can be inferred that there are many determining factors and therefore the method of delivery may be one of several possible factors that affect academic performance and therefore not necessarily a predictive factor.

Positive Effects of AI on Student Performance

Hausman, Rigbi, and Weisburd (2025) examined and collected research data between 2018 and 2024 for over 6,000 classes attended by more than 36,000 students. The researchers found that there was an increase in grades of 1 to 1.5 points in courses where the use of generative AI is allowed. There was also a reduction in the failure rate of

almost 33%. Nevertheless, they have also demonstrated through their results that when using generative AI, the range in which students are graded becomes compressed. Consequently, it can be said that AI may result in little difference being made based on individual differences of students.

A meta-analysis performed by Hu, Pang, & Xing (2025) evaluated twenty-one independent studies and forty-four separate effects found a "moderately" positive effect size regarding the impact of AI on both cognitive and affective/behavioral student outcomes. It should be noted that the effect size for this analysis was .572. A further evaluation of these analyses was then compared to an evaluation of fifty-seven different higher education settings. In comparison to the overall evaluation of the use of AI, the evaluations of the higher education settings had a significantly larger effect size (.684). This indicates a substantially greater improvement than what would be considered to be moderate. This suggests that the potential benefits of utilizing AI in education will be present regardless of the educational setting and/or type of material presented to the students.

Negative Effects of AI on Student Performance

Studies from 2015 through 2025 that involved a total of 32 empirical studies regarding AI chatbots for programming have been subject to meta-analyses. The results of these meta-analyses suggest that students learn better with shorter AI chatbot sessions as opposed to longer ones due to the fact that novelty wears off after some time (Wu & Yu, 2024). These findings indicate that using AI chatbots does not always positively impact student learning when they are used in an unstructured manner; therefore, their use should be strategic.

An analysis of the contents of 100 academic articles from 2010-2020 indicates that there will be a number of issues associated with the integration of AI into educational environments including but not limited to: misuse of AI technology, changes to how teachers interact with their students and/or vice versa, and the potential for unethical uses of AI technologies. Therefore, all parties involved in integrating AI into educational settings need to take responsibility to integrate AI in a responsible and ethically-based manner that supports student learning rather than undermines it (Zhai et al., 2021).

Mixed Findings

The authors of one study examined how well ChatGPT-4, ChatGPT-3.5, Google’s Bard, and Microsoft’s Bing performed on the same college entrance exams for first-year

college students. While all four artificial intelligence systems scored an average of what first-year college students had achieved in their classes, they failed miserably when answering questions with images or requiring cognitive thinking; meanwhile the students' test results did not vary much (Rai et al, 2025). In this way, being aware of the artificial intelligence systems' abilities will help teachers make sure that their tests are written appropriately for each system.

Additionally, Gubanova conducted a survey of high school students regarding whether or not they used artificial intelligence tools during their studies. She determined that most high school students do not believe that artificial intelligence is a good tool to use when doing mathematical problems or writing creative essays; however, many find artificial intelligence to be useful in tutoring them in how to correct their grammatical errors and how to locate formulas online (Gubanova, 2025).

Limitations and Gaps

The present body of research indicates that there are a number of very significant gaps. Many studies investigating performance online compared to in-person were conducted prior to the extensive use of Generative AI, which limits the ability to determine if the use of AI is causing any performance difference seen today (Cavanaugh & Jacquemin, 2015; Hoffman & Elmi, 2020). In addition, while many studies have relied upon participant reports of their own behavior or focused solely on one institution to limit the need for generalization in findings, our study attempts to add to the previous work with an investigation of whether the specific use of AI tools will moderate the relationship between course format and academic performance within STEM courses offered at four-year colleges/universities, utilizing objective grades as measured after the time frame when Chat GPT was introduced into everyday life.

Research Design and Methods

In this study, a mixed-methods approach is adopted to investigate the influence of generative AI tools on the perception of performance of STEM undergraduates in online and face-to-face environments. The motivation for this research is to see whether AI is harmful or helpful to students and their education. We define AI to be helpful if it falls within these parameters: improved GPA, creativity, understanding of material, and overall performance and output. We were hoping to gain clarity on whether AI can be a tool such as a calculator for students or whether its use creates a dependency that harms learning. Through this investigation we sought advice and information on helpful

strategies and qualitative data that can be used to support the use of AI or as a suggestion to avoid its use generally. We also seek to gain a greater understanding of why students use generative AI, how they use it, and when. Through these interviews we hope to also understand student motivations to gain a greater understanding of the effects of AI and whether it causes more harm than good. We also wanted to gain more insight on the mental health of students using generative AI regularly, whether they stressed or felt more relieved, did they feel better prepared, or behind on academic material with repeated use?

Participants/Research Site

The study will involve approximately 50 undergraduate students aged between 18 and 25 years old from the University of Washington, located in Seattle, Washington. The University of Washington was selected as the research site because it is a large, public four-year institution with a diverse undergraduate population across both online and in-person STEM course offerings, making it well-suited for comparing learning environments. Participants will include students of any gender from both online and in-person courses across various STEM majors. Those who frequently use generative AI systems such as ChatGPT, Gemini, and Claude for academic reasons can qualify for participation in this research study, since the study aims to reach students who are not yet experts and who are beginning their educational journeys outside of K-12 education. In addition, approximately 10 students will be included in follow-up interviews for further information.

The study will use convenience sampling to recruit participants from a variety of locations including but not limited to college forums, university Discord Channels, social media platforms and/or posting in classrooms. Participants recruited via the aforementioned sources are likely to have an existing comfort level and interest in technology and artificial intelligence. As such, the sample collected may potentially represent a biased representation of the larger undergraduate student body. Further limitations of this methodology will be discussed in the Limitations section. These participants were chosen because college students are one of the target demographics actively using generative AI systems in an academic context. Including both online and traditional college students will facilitate comparison between distinct learning environments in relation to the usage of generative AI systems and academic performance. Based on the qualitative field study method proposed by Wobbrock (2015), a thorough account of participants should be provided along with an explanation of their selection.

Materials and Tools

A survey (online) and semi-structured interviews will collect data. The questionnaire is being developed using Google Forms and includes both multiple-choice questions as well as open-ended questions that assess how often the participant uses AI, what types of AI are used, how AI impacts their ability to complete school-related work, and whether or not the participant believes that AI has impacted their grade level or academic achievement. Participants who choose to conduct their Zoom interview need to have a device with both internet access and audio capability. In-person interviews require a recording device, pen/paper for note-taking, and a pre-determined set of interview questions. The researcher will store all surveys/semi-structured interviews securely in a password-protected Google Drive document(s) available only to the researcher.

Interview and Survey Questions

These are the sample questions we can use as a basis to collect our research data. The survey questions consist of:

(1) How often do you currently utilize generative AI tools (ChatGPT, Gemini, etc.) for educational purposes?

(Never / Rarely / Occasionally / Frequently / Almost always)

(2) What generative AI tools have you utilized most often during your coursework? (Check all that apply)

(3) For which educational activities do you utilize generative AI?

(e.g. Brainstorming, Writing drafts, Summarizing texts/Readings, Studying, Checking Grammar & Punctuation, Solving Problems)

(4) Primarily do you take courses online, in-person, or a combination of both?

(5) As a result of utilizing these AI tools routinely, has your grade point average increased, remained unchanged or decreased?

(Increase / Remains the Same / Decrease / I am Not Sure)

(6) To what degree do you agree that utilizing generative AI tools has positively impacted your comprehension of course content?

(Scale 1 = Strong Disagree; Scale 5 = Strong Agree)

(7) On how many occasions do you find yourself feeling reliant on generative AI to perform an assignment you would normally complete on your own?

(8) Do you believe utilizing generative AI tools has negatively affected your self-confidence in your academic abilities? If yes, please explain.

The semi-structured interviews also provide guidance and include:

(1) Can you tell me how you usually go about using AI tools within your coursework?

(2) Have you noticed any changes in either your academic performance or your ability to grasp course concepts after beginning to use AI? If so, can you relate one example.

(3) Do you perceive that the utilization of AI differs depending upon whether a class is taught face-to-face versus online? In what ways does this impact your experience?

(4) Are there times when using AI leaves you feeling more anxious or less certain of your academic abilities than you did previously.

(5) Do you ever find that you are unable to complete an assignment without utilizing some type of AI tool. How do you rationalize that?

(6) Do you believe that either your instructor(s), or the institution at large views your usage of AI as acceptable or unacceptable? Does their view impact how you choose to use AI?

Design and Methodology

This research project employs a mixed-methods research design that integrates both quantitative survey data and qualitative interview data. The mixed methods approach was chosen because it allows researchers to identify broader trends in AI usage across a larger sample while also capturing detailed individual accounts of student experiences and perceptions. As stated by Wobbrock (2015), qualitative field studies can be very helpful in understanding participants' behavior in more detail.

The main independent variable for this study is the frequency and nature of generative AI usage among college students. The dependent variable in this research is perceived academic performance. For purposes of this study, it was operationalized as (1) self-reporting (reporting on personal experiences) that reflects changes in GPA and course pass rate; (2) Subjective perception of a student's ability to learn efficaciously, to develop

effective study habits, and to develop an increased level of confidence in managing coursework. Researchers will also analyze differences between students who take primarily online courses and those who take primarily in-person courses.

A semi-structured interview technique that is not randomized and conducted within local environments was chosen since this method provides for flexibility in interviews but does not lose focus on the research question asked by researchers at the same time. Randomized interviews are not realistic due to a lack of rapport/length of interviews would be more plausible with other methods such as surveys. Surveys were used for data collection since this technique allows gathering more data from more respondents quickly.

Procedure

First, participants will be recruited through a message that includes details about the study's purpose and a link to the online survey. Prior to filling out the survey, participants will read an informed consent document explaining the purpose of the study, its voluntary nature, and how confidentiality will be maintained. Participants must accept the consent form before proceeding.

One important concern when conducting this study is asking participants to disclose sensitive information regarding their academic performance. Many students may find it difficult to disclose such personal academic history which could include self-reported GPA and/or individual course grades. In previous studies, researchers have found that asking participants to provide a record of their academic performance on an uncompensated basis and/or providing no apparent privacy protection can result in significant reductions in participant response and create self-selection biases. Specifically, those students who have lower GPAs tend to respond at much higher levels than those with higher GPAs. Therefore, in order to address this issue all GPA-related questions will ask participants to report how they perceive their grades changed during the time period studied. Additionally, prior to collecting any data from participants, participants will be informed that the data they provide will be kept confidential and anonymous. Further, participants will be told that the data they collect will not be released to either instructors or the University. Finally, participants will be provided with a detailed description of the purpose of collecting the data.

Once consent is provided, participants will complete the survey, which should take approximately 10 to 15 minutes. Upon completion, participants will have the option to sign up for a follow-up interview. Researchers will reach out to all those people interested in participating through email to set up a 20-30 minute interview either online

using Zoom, or in a library study room located in the area preferred by the participant. The interviews were designed as a method to collect information from participants in addition to what the survey can capture (open-ended narrative description of emotional response to AI, and detailed descriptions of how they have used their computer with AI). All participants who have only completed the survey will be able to make a full contribution to the statistical/quantitative analysis of this research.

All interview sessions will be audio-recorded with participant consent. Participants will be asked about their use of AI tools in coursework, whether AI has affected their grades or learning, and how their experiences differ depending on whether they study online or in person. After each interview, audio recordings will be transcribed for analysis. All personal identifiers will be replaced with pseudonyms throughout the process to ensure anonymity. No compensation will be provided, as compensation may introduce bias and pressure participants into participating or alter the honesty of their responses.

Quantitative Analysis

Survey data will be analyzed using descriptive statistics to identify trends in AI usage and perceived academic performance. Both researchers will compute means and percentages for responses related to frequency of AI use, types of tools used, and self-reported impact on grades and learning. Responses from students in online courses will be compared to those in in-person courses to identify differences in AI usage patterns. Data will be organized and visualized using Google Sheets, with bar graphs and charts used to present key trends clearly.

Qualitative Analysis

Interview transcripts and open-ended survey responses will be analyzed using thematic analysis. Both researchers will independently code transcripts using open coding, identifying important ideas and repeated themes in participant responses. Codes will then be grouped into broader themes such as AI as a study tool, dependency concerns, academic honesty, effects on critical thinking, and differences between online and in-person learning experiences. Once both researchers have coded independently, they will compare and discuss their codes to reach a consensus on final themes. According to Wobbrock (2015), coding and theme creation are key processes in qualitative analysis in field research studies. Any outliers or participants with unique circumstances will be noted separately and discussed in the findings.

Discussion

While data collection and analysis are still pending, we anticipate that findings will reflect a mixed nature. Some students will report improved academic performance with AI use while others will report no meaningful change or even negative effects on their learning. More importantly, we expect qualitative interviews to reveal nuances that survey data alone cannot capture, such as how students emotionally relate to AI, whether they feel dependent on it, and how their experiences differ based on their learning environment (online vs. in-person). This work is important because it centers the student perspective at a time when institutions are making major policy decisions about AI without fully understanding how students are actually using these tools and what effects that use is having on their education and overall well-being.

Limitations

Several limitations should be noted. First, the study relies on convenience sampling through university forums and social media, which may attract participants who are already comfortable with or interested in AI, potentially skewing results toward more positive perceptions. Second, the study is limited to undergraduate students at the University of Washington, limiting generalizability to other institutions and demographics. Third, self-reported grade changes and perceptions of academic performance may not accurately reflect actual GPA outcomes. Finally, since AI tools are evolving rapidly, findings from this study may become outdated quickly as new models and institutional policies emerge.

Future work

The best future study will likely consist of longitudinal studies that follow the same students through more than one semester over more than one institution in other parts of the country at different types of institutions (i.e., smaller liberal arts colleges, community colleges, rural/undeserved college settings). Such studies would enable us to better understand how long-term use of AI may affect an individual's overall academic performance and help determine if our findings are representative of a very large, research-based public university (like UW) or represent something else. The next generation of studies on AI use can expand beyond STEM to include the academic performance of students studying social science, humanities and professional subjects; this will allow us to see if the patterns we identified for STEM students are consistent with those of all undergraduate students.

Conclusion

As generative AI becomes increasingly prominent in higher education, understanding its impact on student learning is more important than ever. This study contributes to that understanding by combining quantitative survey data with qualitative student interviews, which offers a more complete picture than large-scale grade analyses alone. Through the analysis of STEM undergraduates across online and in-person formats, we hope to provide actionable findings for educators and institutions navigating the challenges of integrating AI responsibly. Ultimately, the goal of this work is not to determine with black and white metrics whether genAI is ultimately good or bad for education, but to better understand the conditions under which it supports or undermines genuine learning.

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